Count the Overlap Once

AP Statistics · Unit 2 · Topic 2.5 · TEACHER KEY

CED TETHER

- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-4** — The likelihood of a random event can be quantified.
- **LO VAR-4.C** — Explain why two events are (or are not) mutually exclusive. [Skill 4.B]
- **EK VAR-4.C.1** — The probability that events A and B both will occur, sometimes called the joint probability, is the probability of the intersection of A and B, denoted $P(A \cap B)$.
- **EK VAR-4.C.2** — Two events are mutually exclusive or disjoint if they cannot occur at the same time. So $P(A \cap B) = 0$.

Essential question: How does an intersection determine whether two events are mutually exclusive and how their union should be counted?

DOK-3 skill: 4.B · **FRQ pattern:** `mutually-exclusive-or-not-when-the-addition-rule-double-counts`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Uses the nonzero 35-student overlap to reject mutual exclusivity and support Noor
- **reasoning-2** (required) — Explains double counting and gives 105/120
- **reasoning-3** (required) — Explains that reporting everyone hides 15 students who like neither

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Assuming two named food preferences must be mutually exclusive
- Replacing a computed value above 1 with 1 instead of diagnosing the counting error
- Subtracting the 35 twice, producing 70/120
- Calling ‘likes pizza’ and ‘does not like tacos’ mutually exclusive even though 45 students satisfy both

[STAR] THE PROBLEM — annotated

Suppose 120 students answer two yes-or-no questions: "Do you like pizza?" and "Do you like tacos?" The table summarizes their answers. Let P be the event that a randomly selected student likes pizza and T the event that the student likes tacos.

Jamie: "$P(P \text{ or } T) = 80/120 + 60/120 = 140/120$. Because a probability cannot exceed 1, this must mean everyone likes at least one; I would report the answer as 1."

Noor: "Some students may have been counted in both totals, so we need the intersection before deciding."

Student food preferences (counts)

Preference	Pizza	Not pizza	Total
Tacos	35	25	60
Not tacos	45	15	60
Total	80	40	120

FIRST TAKE — one sentence before you work the parts

Does the table support reporting everyone likes pizza or tacos? Choose a claim and cite one count.
Key: Not graded. A student may initially trust the marginal totals; the finish should make the duplicated intersection visible.

[BOOK] INTERSECTIONS AND MUTUALLY EXCLUSIVE EVENTS (keep on the page)

$A \cap B$ is the intersection: outcomes in both events. Events are mutually exclusive (disjoint) when they cannot happen together, so $P(A \cap B) = 0$. For a random selection from equally likely individuals, divide the number in the event by the total. Count each individual once. The complement rule is $P(A^c) = 1 - P(A)$. Every probability must be between 0 and 1.

DOK 1 **(a)** State the number in $P \cap T$ and write its probability as a fraction.

Key: The intersection contains 35 students, so $P(P \cap T) = 35/120 = 7/24 \approx 0.292$.

DOK 2 **(b)** Find the probability that a selected student likes at least one food, counting each student once. Also find the probability of liking neither.

Key: There are $35 + 25 + 45 = 105$ students who like at least one food, so the probability is $105/120 = 7/8 = 0.875$. There are 15 who like neither, with probability $15/120 = 1/8 = 0.125$.

DOK 3 **(c)** Are the two events mutually exclusive, and whose reasoning is supported? Use the overlap and your probability to explain Jamie's error. Explain what reporting everyone would hide. Write 3–4 sentences.

Key: The events are not mutually exclusive because 35 students like both foods, so Noor's reasoning is supported. Jamie counted those 35 twice: $80 + 60 - 35 = 105$ students like at least one, giving $105/120 = 0.875$. Reporting everyone would hide the 15 students who like neither; a probability above 1 signals a counting error.